Week 4 — Linux Systems Administration

Overview

This week provided hands-on Linux systems administration experience in a lab environment. Linux is the dominant OS in server infrastructure, cloud environments, and security tooling — this week built the command-line fluency that underpins all later technical work.

Topics Covered

- Linux filesystem hierarchy and navigation
- User and group management
- File permissions and access control
- Package management (`apt`)
- Process management
- Scheduling tasks with `cron`
- System hardening fundamentals

Assignment

Completed a series of systems administration tasks on a Linux VM, simulating the responsibilities of a junior sysadmin or security engineer:

- Created and managed user accounts and groups
- Configured file permissions using `chmod` and `chown`
- Locked down sensitive files and directories
- Set up automated tasks using `cron`
- Investigated running processes and system resource usage

- Applied basic system hardening steps (disabling unused services, managing sudo access)

Key Commands & Concepts

User & Group Management

```bash
# Create a new user
sudo adduser username

# Add user to a group
sudo usermod -aG groupname username

# View user info
id username
cat /etc/passwd
```

File Permissions

```bash
# Change permissions (octal notation)
chmod 640 filename      # Owner: rw, Group: r, Others: none

# Change ownership
chown user:group filename

# View permissions
ls -la
```

Cron Scheduling

```bash
# Edit crontab
crontab -e

# Example: run script every day at 2am
0 2 * * * /path/to/script.sh
```

Process Management

```bash
ps aux          # View all running processes
top / htop      # Real-time process monitor
kill -9 PID     # Force-terminate a process
```

Hardening Principles Applied

- Principle of Least Privilege — users only get the access they need
- Disabling root SSH login
- Removing or disabling unused services
- Auditing sudoers file
- Reviewing world-writable files and SUID/SGID binaries